When are consecutive and Single-Step reactions distinguishable ?

$k_1 >> k_2$

$$[A] = [A]_0 e^{-k_1 t}$$

$$[P] = (1 - \exp(-k_2 t))[A]_0$$

Rate of formation of product is different from rate of decay of A

$k_1 << k_2$

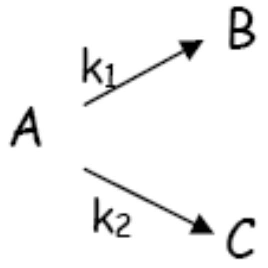
$$[A] = [A]_0 e^{-k_1 t}$$

$$[P] = (1 - \exp(-k_1 t))[A]_0$$

Rate of formation of product is same as rate of decay of A

Parallel Reaction

Parallel 1st order reactions



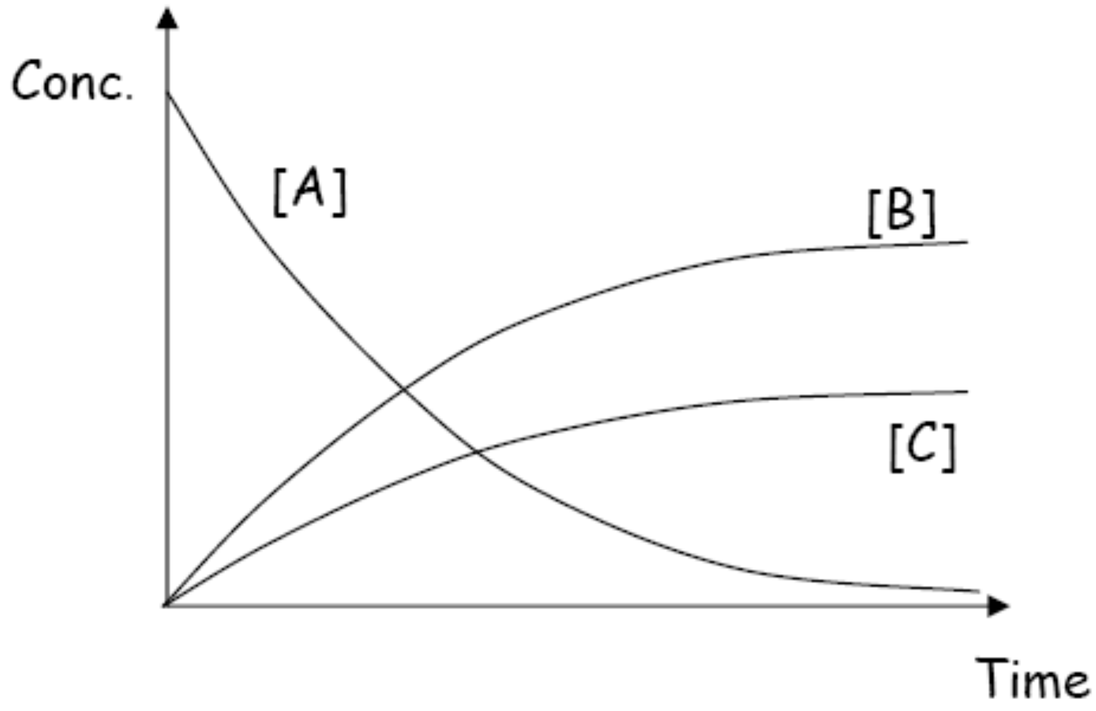
$$\frac{-d[A]}{dt} = k_1[A] + k_2[A]$$

$$[A] = [A]_o e^{-(k_1+k_2)t}$$

$$[B] = \frac{k_1[A]_o}{k_1 + k_2}\left(1 - e^{-[k_1+k_2]t}\right)$$

$$[C] = \frac{k_2[A]_o}{k_1 + k_2}\left(1 - e^{-[k_1+k_2]t}\right)$$

Parallel Reaction



Reversible Reaction

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B \qquad\qquad K_{eq} = \frac{[B]_{eq}}{[A]_{eq}}$$

1^{st} order reversible reactions

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B \qquad\qquad -\frac{d[A]}{dt} = k_1[A] - k_{-1}[B]$$

$$[B] = [B]_o + ([A]_o - [A])$$

$$\text{So...} \quad -\frac{d[A]}{dt} = k_1[A] - k_{-1}([B]_o + [A]_o - [A])$$

Reversible reaction

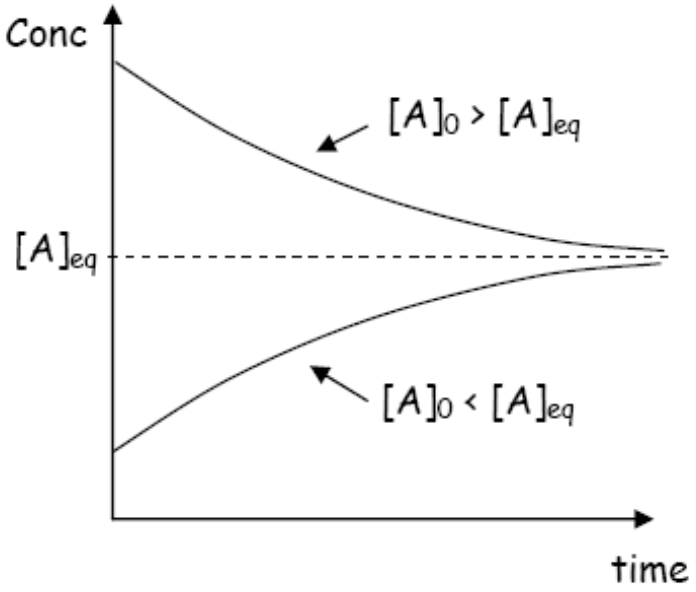
At Equilibrium, $\frac{d[A]}{dt} = 0$

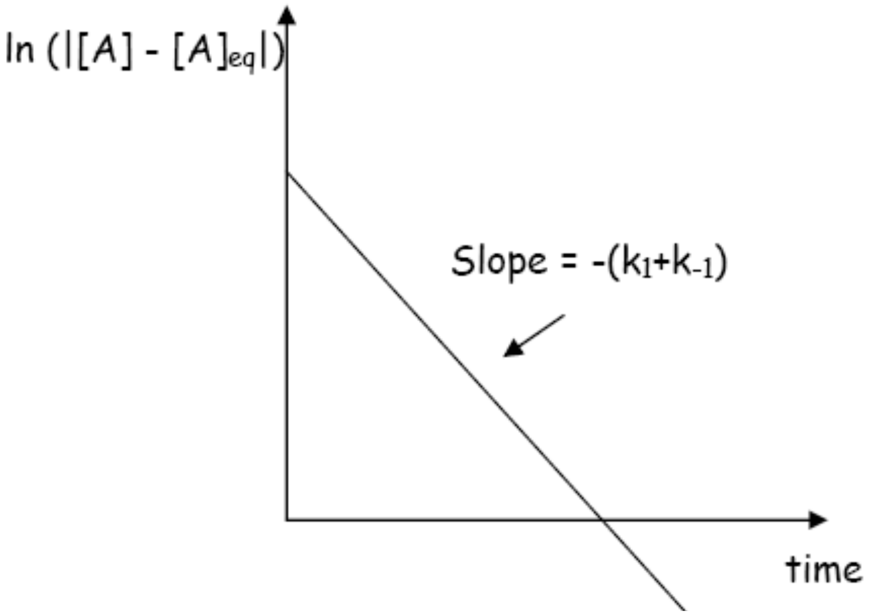
$$\Rightarrow \quad \boxed{[A]_{eq} = \frac{k_{-1}}{k_1 + k_{-1}}\left([B]_o + [A]_o\right)}$$

$$-\frac{d\left([A] - [A]_{eq}\right)}{dt} = -\frac{d([A])}{dt} = \left(k_1 + k_{-1}\right)\left([A] - [A]_{eq}\right)$$

$$\Rightarrow \quad \boxed{[A] - [A]_{eq} = \left([A]_o - [A]_{eq}\right)e^{-(k_1 + k_{-1})t}}$$

Reversible reaction





<u>Can measure:</u> $K_{eq} = \dfrac{k_1}{k_{-1}}$ and $k_1 + k_{-1} \equiv k_{obs}$

And extract k_1 <u>and</u> k_{-1}

Determining Mechanism from Rate Law

- If the rate law is $r = k[A]^{\alpha}[B]^{\beta}.........[L]^{\lambda}$, the total composition of the reactants in the rate limiting step is αA+βB+….. $+ nH_2O$

- If the rate law is $r = k[A]^{\alpha}[B]^{\beta}.........[L]^{\lambda}/[M]^{\mu}[N]^{\nu}$, the total composition of the reactants in the rate limiting step is αA+βB+…..-μM-νN…….. $+ nH_2O$

- A rate law, to be properly interpreted according to rule 1, must be written in terms of the predominate species in the reaction medium.

Mechanism

- The concentration dependence in rate law establish the elemental composition of Transition state and its charge.

Determining Mechanism from Rate Law

- A summation of n terms in the denominator implies a succession of n steps, where at least one step is reversible.
- Species whose concentrations appear in single-term denominators are produced in the step prior to rate controlling step.
- If there are three terms in the denominator, intermediate can disappear in three or more steps.

Determining Mechanism from Rate Law

- The number of positive terms in the rate law is the number of independent, parallel pathways. Negative terms represent the reverse reaction.
- Adding up the steps in a mechanism must yield the net chemical reaction; rapid reaction may follow the rate-controlling step.
- Alternative mechanisms leading to same pattern of activated complexes are not kinetically distinguishable.
- The order increases with increasing concentration when the reaction proceeds by parallel pathways but decreases when a series of steps occur.